

Trainings ▾

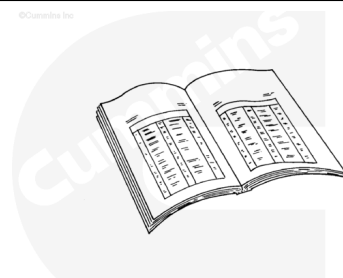
Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environment regulations.



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Note : The aftertreatment fuel injector fuel and coolant lines may have enough flexibility to allow for the removal of the aftertreatment adapter pipe without removing the aftertreatment fuel injector.

- Remove the two aftertreatment intake NOx sensor metal sensor body from the adapter pipe. See equipment manufacturer service information.

Note : If removal of the aftertreatment fuel injector is needed, determine the appropriate radiator or coolant expansion tank cap adapter for the vehicle being serviced. A list of available caps and applications can be found in Section 14 of the Service Products Catalog, Bulletin 3377710.

Use a vacuum pump type device on the cooling system. Install the adapter on the radiator or coolant expansion tank. Connect an automotive hand-held vacuum pump and apply 3 in Hg [40 in H₂O] of vacuum.

Note : Depending on the cooling system configuration, more vacuum may be required to keep the cooling system in balance.

Place a container under the aftertreatment fuel injector. There may be a small quantity of coolant that drains from the coolant lines when disconnected.

- Disconnect the aftertreatment injector and coolant lines. Refer to Procedure 011-051 in Section 11.
(/qs3/pubsys2/xml/en/procedures/132/132-011-051-tr-vqxi6.html)
- Disconnect the electrical connection.
- Remove the aftertreatment injector from the adapter pipe. Refer to Procedure 011-042 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr-mixar5.html)

Remove

Remove the V-band clamp between the exhaust adapter pipe and the turbocharger.

Remove the V-band clamp between the exhaust adapter pipe and the OEM exhaust down-pipe.

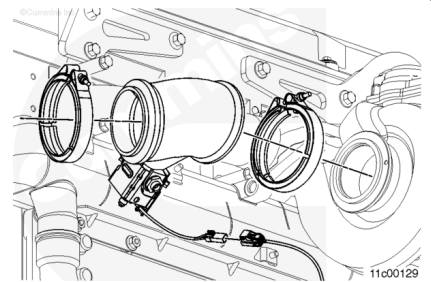
Remove the adapter pipe assembly from engine.

Remove the gasket from each end and discard. Do **not** use a screwdriver or other sharp tool to remove the gaskets. This will damage the gasket seating area.

Remove the aftertreatment injector from the adapter pipe.

Refer to Procedure 011-042 in Section 11.

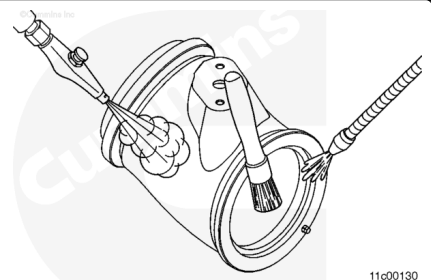
(/qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html)



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Saturate the inside of the adapter pipe with safety solvent.

Mineral based solvents can be used if results from safety solvent are not acceptable.

Clean the inside of the adapter pipe with a non-abrasive brush.

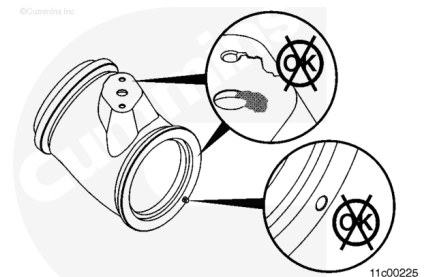
Saturate the injector mounting pad with safety solvent.

Clean the mounting surface with a non-abrasive brush.

Dry with compressed air.

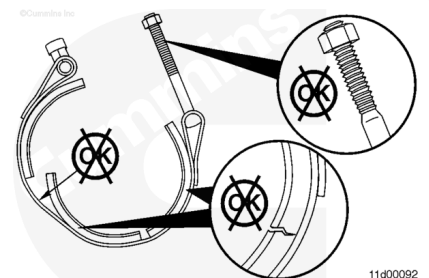
Make sure the locator pin is in the adapter pipe. Replace the adapter, if necessary.

Inspect all mating surfaces for cracks or any other damage. Replace the adapter, if necessary.



Inspect the V-band clamps for signs of overextension. The band must not be bent or cracked.

Replace the V-band clamp if damage is found.



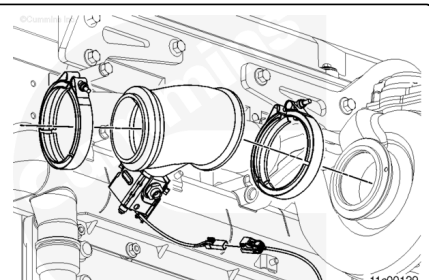
Install

Install the aftertreatment injector assembly onto the adapter pipe. Refer to Procedure 011-042 in Section 11.

(</qs3/pubsys2/xml/en/procedures/101/101-011-042-tr.html>)

Install new gaskets on each end of the adapter pipe.

Place the exhaust adapter pipe in position with the pin in the correct location in the turbocharger.



Tighten the V-band clamps on each end.

Torque Value: 11 n•m [97 in-lb]

Finishing Steps

- Connect the electrical connection.
- Connect the aftertreatment injector coolant lines. Refer to Procedure 011-051 in Section 11.
(/qs3/pubsys2/xml/en/procedures/101/101-011-051-tr.html)
- If the coolant was drained, fill the cooling system.
 - For Signature™, ISX, and QSX15 engines, use the following procedure in the Signature™, ISX, and QSX15 Service Manual, Bulletin 3666239. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
 - For ISM, ISMe, and QSM11 engines, use the following procedure in the ISM, ISMe, and QSM11 Service Manual, Bulletin 3666322. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/35/35-008-018-tr.html)
- If a vacuum pump was used, remove the vacuum pump.
- Start the engine and check for leaks.

